Xiaoyang Xu

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Education

University of California, Santa Barbara Sep. 2024–present
Ph.D. student in Computer Science; advisor: Daniel Lokshtanov.

University of Science and Technology of China Sep. 2020–Jun. 2024
B.Eng. (Honors) in Computer Science, School of the Gifted Young.

Selected Honors and Awards

Outstanding Poster Award, ICBS College Poster Session; *Efficiency of Pairing Heaps*. 2026
GUO Moruo Scholarship, the highest honor at USTC. 2024
Gold Award, ICPC China National Invitational Programming Contest (Kunming). 2024
China National Scholarship, Ministry of Education. 2023
Selected Participant, 10th Heidelberg Laureate Forum. 2023

Service

Sub-reviewer: SODA 2027 (2×), SOSA 2027, IWOCA 2026, ITCS 2026, ESA 2025 (2×).
Co-coach, UC Santa Barbara Competitive Programming Teams. Nov. 2024–present
Teaching assistant:

- UCSB CMPSC 130A: Data Structures and Algorithms I (Spring 2026, Fall 2024).
- UCSB CMPSC 132: Algorithmic Engineering (Fall 2025).
- UCSB CMPSC 130B: Data Structures and Algorithms II (Spring 2025, Winter 2025).
- USTC 011186: Algorithms for Big Data (Spring 2023).

Academic Visits

New York University Shanghai, with Jie Xue. Sep. 2026
University of Bergen, with Fedor V. Fomin. Feb.–Mar. 2026
Toyota Technological Institute at Chicago, with Santhoshini Velusamy. Jun.–Sep. 2025
Harvard University, with Heng Yang. Jul. 2023–Jan. 2024

Publications

[1] **Efficiency of the Pairing Heap.**
Xiaoyang Xu, Robert E. Tarjan. *SODA 2027*, to appear.

[2] **Pure Pairing Heaps.**
Robert E. Tarjan, **Xiaoyang Xu**. In submission, 2026.

[3] **Learning-Augmented Streaming Algorithms for Approximating Boolean Max-CSPs.**
Yinhao Dong, Pan Peng, Ali Vakilian, Santhoshini Velusamy, **Xiaoyang Xu**. *NeurIPS 2026*, to appear.

[4] **Parameterized Approximation of Rectangle Stabbing.**
Huairui Chu, Ajaykrishnan E S, Daniel Lokshtanov, Anikait Mundhra, Thomas Schibler, **Xiaoyang Xu**, Jie Xue. *ESA 2026*.

[5] **Fast and Certifiable Trajectory Optimization.**
Shucheng Kang, **Xiaoyang Xu**, Jay Sarva, Ling Liang, Heng Yang. *WAFR 2024*.

[6] **A Novel Skip Orthogonal List for Dynamic Optimal Transport Problem.**
Xiaoyang Xu, Hu Ding. *AAAI 2024*.

Talks and Posters

Efficiency of Pairing Heaps

- Theory Lunch, New York University Shanghai, September 16, 2026.
- ICBS 2026 College Poster Session. [Poster]

Learning-Augmented Streaming Algorithms for Approximating Max-DICUT

- Algorithms Seminar, University of Bergen, March 12, 2026. [Slides]

Skip Orthogonal List for Dynamic Optimal Transport

- AAAI Conference on Artificial Intelligence, 2024. [Poster]